

1. Why CSS? Inline vs. Internal vs. External

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Local Community Event Portal</title>

  <!-- Internal (embedded) style -->
  <style>
    body {
      background-color: #f4f4f4;
    }
  </style>

  <!-- External stylesheet -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <!-- Inline style -->
  <h1 style="color: red;">Welcome to the Community Event Portal</h1>

  <h2>Upcoming Events</h2>

</body>
</html>
```

styles.css

```
/* Header styles */
h1 {
  font-size: 32px;
}

/* Body/reset styles */
body {
  margin: 0;
  padding: 0;
}
```

This shows all three CSS methods: inline (on the h1), internal (the `<style>` block in the head for the background), and external (the linked `styles.css` file, which is where reusable styles should generally live).

2. CSS Syntax and Comments

styles.css (continued)

```
/* =====
   Base Reset
   ===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

/* =====
   Navigation Styles
   ===== */
nav {
  background-color: #333;
  padding: 10px 20px;
}

/* =====
   Call To Action Button
   ===== */
/* Style for main CTA button */
.cta-button {
  background-color: #007BFF;
  color: white;
  padding: 10px 20px;
  border: none;
  border-radius: 4px;
  cursor: pointer;
}

.cta-button:hover {
  background-color: #0056b3;
}
```

Good CSS hygiene here: consistent 2-space indentation, one rule per line, and comment banners that section off the file so anyone joining the project can scan it quickly.

3. Selectors Playground

```
/* Universal selector - reset margin/padding */
```

```
* {  
  margin: 0;  
  padding: 0;  
}
```

```
/* Element selector - all h2 headings */
```

```
h2 {  
  color: #2c3e50;  
  font-size: 24px;  
}
```

```
/* ID selector - the main banner */
```

```
#mainHeader {  
  background-color: #007BFF;  
  color: white;  
  padding: 20px;  
  text-align: center;  
}
```

```
/* Class selector - event containers */
```

```
.eventCard {  
  background-color: #fff;  
  border: 1px solid #ddd;  
  border-radius: 6px;  
  padding: 15px;  
  margin-bottom: 15px;  
}
```

```
/* Grouping selector */
```

```
h3, p {  
  font-family: Arial, sans-serif;  
  line-height: 1.5;  
}
```

Matching HTML

```
<header id="mainHeader">  
  <h1>City Events Hub</h1>  
</header>
```

```
<div class="eventCard">  
  <h3>Summer Music Festival</h3>
```

```
<p>Join us in the park this Saturday for live music and food trucks.</p>
</div>
```

4. Color & Background Styling

```
/* HEX and RGBA colors */
```

```
body {
  color: #222222;
  background-color: #f9f9f9;
}
```

```
.eventCard {
  background-color: rgba(0, 123, 255, 0.1);
  color: rgba(0, 0, 0, 0.85);
}
```

```
/* Background image with fallback color */
body {
  background-color: #cccccc; /* fallback if image fails to load */
  background-image: url("images/city-banner.jpg");
  background-size: cover;
  background-position: center;
  background-repeat: no-repeat;
}
```

```
/* Gradient section headers */
.section-header {
  background: linear-gradient(to right, #007BFF, #00c6ff);
  color: white;
  padding: 15px;
  text-align: center;
}
```

5. Typography: Fonts and Text

In the HTML head

```
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">
```

Or with `@import` at the top of `styles.css`:

```
@import url('https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap');
```

CSS

```
body {  
  font-family: 'Poppins', Arial, sans-serif;  
}
```

```
h1 {  
  font-size: 36px;  
  font-weight: 600;  
  font-style: normal;  
}
```

```
.eventCard p {  
  font-size: 16px;  
  font-style: italic;  
  text-align: justify;  
  letter-spacing: 0.3px;  
  line-height: 1.6;  
}
```

```
.section-header {  
  text-transform: uppercase;  
}
```

6. Link and List Styling

```
/* Pseudo-classes for links */
```

```
a:link {  
  color: #007BFF;  
  text-decoration: none;  
}
```

```
a:visited {  
  color: #6c757d;  
}
```

```
a:hover {  
  color: #0056b3;  
  text-decoration: underline;  
}
```

```
a:active {  
  color: #003d80;  
}
```

```
/* Navigation list - remove bullets */
```

```
nav ul {
```

```
list-style-type: none;
list-style-position: outside;
margin: 0;
padding: 0;
}
```

```
nav ul li {
  display: inline-block;
  padding: 10px 15px;
  margin-right: 5px;
}
```

```
/* General list styling */
.event-list li {
  list-style-type: square;
  padding: 5px 0;
  margin-left: 20px;
}
```

7. Table Styling

HTML

```
<table class="admin-table">
  <tr>
    <th>Event</th>
    <th>Date</th>
    <th>Location</th>
  </tr>
  <tr>
    <td>Music Festival</td>
    <td>July 15</td>
    <td>Central Park</td>
  </tr>
  <tr>
    <td>Food Fair</td>
    <td>July 22</td>
    <td>Main Square</td>
  </tr>
</table>
```

CSS

```
.admin-table {
  width: 100%;
}
```

```
border-collapse: collapse;
text-align: center;
}
```

```
.admin-table th,
.admin-table td {
border: 1px solid #ccc;
padding: 10px;
}
```

```
.admin-table th {
background-color: #007BFF;
color: white;
}
```

```
/* Zebra striping */
.admin-table tr:nth-child(even) {
background-color: #f2f2f2;
}
```

8. Box Model & Layout Control

```
.eventCard {
border: 2px solid #007BFF;
padding: 20px;
margin: 15px auto;
outline: none;
}
```

```
/* Highlight a selected form field */
input:focus {
outline: 2px solid orange;
}
```

```
/* visibility vs display comparison */
.hidden-but-takes-space {
visibility: hidden; /* element is invisible but still occupies layout space */
}
```

```
.removed-from-layout {
display: none; /* element is invisible and takes up no space at all */
}
```

Open Chrome DevTools (right-click → Inspect → Computed/Box Model tab) on `.eventCard` to see the margin, border, padding, and content box drawn visually, and try dragging the values live to see how spacing changes.

9. Multiple Columns in Text

HTML

```
<section class="bulletin">
  <h2>Community Bulletin</h2>
  <p>
    The city council has announced new plans for the downtown renovation
    project, including expanded green spaces, a farmers market pavilion,
    and improved pedestrian pathways connecting the library to the park...
  </p>
</section>
```

CSS

```
.bulletin p {
  column-count: 2;
  column-gap: 30px;
  column-rule: 1px solid gray;
}
```

10. Responsive Web Design with Media Queries

```
/* Base (desktop-first) styles */
nav ul li {
  display: inline-block;
}
```

```
.eventCard img {
  width: 100%;
  max-width: 400px;
}
```

```
.hero {
  height: 60vh;
  width: 100%;
}
```

```
/* Media query for screens smaller than 768px */
@media (max-width: 768px) {
  nav ul {
```



```
    display: flex;
    flex-direction: column;
  }

  nav ul li {
    display: block;
    width: 100%;
  }

  .eventCard img {
    max-width: 100%;
  }

  h1 {
    font-size: 24px;
  }

  p {
    font-size: 14px;
  }
}
```

Bonus: Flexbox layout for event cards

```
.event-grid {
  display: flex;
  flex-wrap: wrap;
  gap: 20px;
}

.eventCard {
  flex: 1 1 300px;
}
```

11. Debug and Test with Dev Tools and VS Code

This one is a workflow exercise rather than code, so here's what to actually do:

Open Chrome DevTools with F12 or Ctrl+Shift+I (Cmd+Option+I on Mac).

Click the device toolbar icon (or Ctrl+Shift+M) to simulate phones, tablets, and desktops, and test the breakpoint from Exercise 10 by resizing.

Use the Elements panel to click on any component (like `.eventCard`) and see its live CSS rules in the Styles pane; you can toggle properties on/off or edit values directly and watch the page update in real time.

Open the Network tab, reload the page, and confirm `styles.css` shows a 200 status, so you know the external stylesheet actually loaded rather than silently failing.

In VS Code, install the Live Server extension so you can right-click `index.html` and choose "Open with Live Server" — this auto-refreshes the browser every time you save a CSS change, which speeds up this whole debugging loop.

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